30 September 1959

# MILITARY SPECIFICATION

## FIRING DEVICE, DEMOLITION MARK 6 MOD 0

### 1. SCOPE

1.1 Scope. - This specification covers the requirements for the procurement of the Demolition Firing Device Mark 6 Mod 0.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes and appendix thereto.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 414458 - Firing Device, Demolition, Mark 6 Mod 0; List of Drawings, Assemblies, Parts, Specifications, etc.
LD 414462 - Container Assembly for Firing Device, Demolition Mark 6 Mod 0; List of Drawings, Assemblies, Parts, Specifications, etc.
LD 416389 - Shipping Container for 50 Firing Devices, Demolition Mark 6 Mod 0.

FSC 1375

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PUBLICATIONS

BUREAU OF ORDNANCE

OP 400 - General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. - General requirements for the manufacture of the firing device shall be in accordance with Publication OP 400. The firing device shall conform to Drawing LD 414458 and all documents listed thereon, this specification, and the contract or order.

3.2 Preproduction sample. - Unless otherwise specified in the contract or order (see 6.2), a preproduction sample is required (see 4.3.1) for each such contract or order. Samples shall be manufactured using the methods, procedures, and plant location proposed for the production lot. Samples shall be inspected as specified under 4.4, such inspection being for the purpose of determining that the manufacturer's production process will produce firing devices that will meet the requirements of this specification.

3.3 Interchangeability. - All parts having the same part number shall be interchangeable without selection or fitting.

3.4 Corrosion protection. - During the course of manufacture the surfaces of all parts and assembled units shall be protected against corrosion prior to and after assembly.

3.5 Performance requirements. -

3.5.1 Firing pin. -

3.5.1.1 Firing pin release. - With the safety pin removed, an axial force on the base of the firing pin of not less than 4 pounds nor more than 10 pounds shall be required to release the firing pin.



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3.5.1.2 Firing pin retention. - When the safety pin is removed the firing pin shall not become dislodged from its installed position by action of the firing pin spring or through ordinary handling of the firing device.

3.5.2 Safety pin. - A force of not less than 10 pounds nor more than 30 pounds shall be required to withdraw the safety pin.

3.5.3 Diaphragm. - The values of forces required to snap the diaphragm and height and deflection of the diaphragm shall be as specified on the applicable drawing.

3.5.4 Leakage. - The seal at the arming cell sleeve and the seal at the diaphragm of the firing device shall show no leakage when tested in accordance with 4.4.3.2.4.

3.5.5 Sensitivity. -

3.5.5.1 Functioning. - The firing device shall fire the primer when tested in accordance with 4.4.3.1.1.1.

3.5.5.2 Safety. - The firing device shall not fire the primer when tested in accordance with 4.4.3.1.1.2.

3.6 Government furnished property. - The Government shall furnish items as specified in 3.6.1, 3.6.2, and 3.6.3.

3.6.1 Arming cells. - The Government shall furnish one each of Arming Cell Mark 1 Mod 7 and Mark 1 Mod 8 in accordance with Specification MIL-A-18924 for each firing device package specified in 5.1.1.2.

3.6.2 Primers. - The Government shall furnish one M27 Percussion Primer in accordance with Specification MIL-P-2497 for installation in each coupling base for each firing device package specified in 5.1.1.2. The Government shall also furnish the required number of the above primers necessary to conduct the tests of 4.4.3.1.1.1 and 4.4.3.1.1.2.

3.6.3 Blasting caps. - The Government shall furnish the required number of J-2 Electric Special Blasting Caps in accordance with Specification MIL-C-14003 necessary to conduct the tests of 4.4.3.1.1.1 and 4.4.3.1.1.2.

3.7 Workmanship. - Workmanship shall be of such quality as to produce firing devices of uniform quality and as to meet all requirements of the applicable drawings and of this specification. Each firing device shall be free from chips, dirt, grease, and other foreign material.



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## 4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Contractor's inspection. - Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Lot. -

4.2.1 For periodic tests. - For the purpose of sampling, a lot for periodic tests shall consist of 500 or more firing devices manufactured by the same process and at the same plant location.

4.2.2 For lot acceptance tests. - For the purpose of sampling, a lot for lot acceptance tests shall consist of all firing devices manufactured by the same process and at the same plant location and submitted for Government acceptance at one time.

4.3 Sampling. -

4.3.1 For preproduction inspection. - Unless otherwise specified (see 4.3.1.1), ten preproduction sample firing devices shall be supplied by the contractor as specified in 3.2. The sample shall be submitted to the Government inspector for preproduction inspection (see 4.4.3.1) at an activity designated in the contract or order (see 6.2) or by the Bureau of Ordnance. Further production of firing devices prior to the approval of the procuring activity or completion of inspection of the preproduction sample shall be at the contractor's risk.

4.3.1.1 Preproduction sample and inspection for a subsequent contract. - If a contractor has previously delivered firing devices in accordance with the requirements of this specification and his product has been found to be satisfactory, the preproduction sample and inspection for any subsequent contract or order may be waived at the discretion of the procuring activity (see 6.2)

4.3.2 Sampling for periodic tests. - Sample size for the periodic tests shall be determined in accordance with Standard MIL-STD-105, Appendix, Inspection Level L8, minimum sample size 15 units per requirement. Acceptance criteria shall be determined in accordance with an AQL of 1.0 percent defective for functional and no defective units permitted for safety test. The frequency of sampling shall be as specified by the procuring activity (see 6.2). Samples to be subjected to the periodic tests as specified in 4.4.3.1 shall be taken from acceptance lots (see 4.2.2) whose samples (see 4.3.3) have passed the lot acceptance tests of 4.4.3.2.



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The Government inspector shall forward the samples to a testing activity designated in the contract or order (see 6.2) or by the Bureau of Ordnance.

4.3.3 Sampling for lot acceptance. - Unless otherwise specified in the contract or order (see 6.2), sampling for lot acceptance inspection of firing devices shall be in accordance with Standard MIL-STD-105, acceptable quality level (AQL) 1.0 percent defective.

4.3.4 Sampling of firing device containers. - A random sample of firing device inner containers (see 5.1.1) and outer shipping containers (see 5.1.1.2) for the inspection of 4.4.2.1 shall be selected by the Government inspector in accordance with the provisions of Standard MIL-STD-105, Inspection Level I, AQL 4.0 percent defective.

4.4 Inspection. -

4.4.1 Government inspection. - The Government representative will examine the contractor's records and processes and will witness or perform examinations and tests as are deemed necessary to assure that components and assemblies of the firing device are in accordance with all applicable requirements.

4.4.2 Visual and dimensional examination. - Firing devices shall be visually examined for completeness of manufacture, freedom from defects, proper identification, and workmanship. Dimensional and other nondestructive inspections shall be made in accordance with the acceptance inspection instructions provided by the procuring activity.

4.4.2.1 Packaging, packing, and marking. - Firing device containers selected in accordance with 4.3.4 shall be examined for proper fill and to determine that packaging, packing, and marking are in accordance with the requirements of Section 5.

4.4.3 Tests and test procedures. -

4.4.3.1 Preproduction and periodic tests. - Preproduction tests shall consist of the tests of 4.4.3.1.1.1 and 4.4.3.1.1.2 and the lot acceptance tests of 4.4.3.2.1.1 through 4.4.3.2.4. The lot acceptance tests shall be conducted prior to the tests of 4.4.3.1.1.1 and 4.4.3.1.1.2. Periodic tests shall consist of the tests of 4.4.3.1.1.1 and 4.4.3.1.1.2. Any sample firing device selected for periodic tests that has not been subjected to the lot acceptance tests of 4.4.3.2.1.1 through 4.4.3.2.4 shall receive these tests prior to the tests of 4.4.3.1.1.1 and 4.4.3.1.1.2 either at the contractor's plant or at the location of periodic testing. Preproduction and periodic samples that



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meet the requirements of this specification shall be included in the quantity called for in the contract or order.

4.4.3.1.1 Sensitivity tests. - If the tests of 4.4.3.1.1.1 and 4.4.3.1.1.2 are successful and the firing devices tested meet the requirements of this specification, the diaphragm of each of these firing devices, having been snapped during the tests, shall be removed and discarded and a new diaphragm installed after each of the above tests. After the test of 4.4.3.1.1.1 has been completed the coupling base of each firing device, due to firing of the primer, shall be discarded and replaced with a new one.

4.4.3.1.1.1 Functioning test. - A suitable open top wooden tank with inside dimensions of 48 inches length, 24 inches width, and 18 inches depth shall be provided by the testing activity. The tank shall contain water to a depth of 12 inches for each firing device tested, the firing devices being tested one at a time. Each firing device to be tested shall have a coupling base containing a primer (see 3.6) installed, shall have the safety pin removed, and shall be without an arming cell (see 3.6). A special electric blasting cap (see 3.6) shall be lowered in the water to a depth of 7 inches. The firing device shall be lowered in the water to a depth of 7 inches and suspended at this level by a metal strap. The minimum distance between the blasting cap and the diaphragm end of the firing device shall be 5 inches. The blasting cap shall be fired and the firing device examined to determine conformance to the requirement of 3.5.5.1. The firing device shall be thoroughly dry before removing the coupling base and before replacing the diaphragm.

4.4.3.1.1.2 Safety test. - The test equipment, preparation of the firing device, and all procedures shall be as specified in 4.4.3.1.1.1 with the exception that the safety pin shall be installed and the distance in the water between the blasting cap and the diaphragm end of the firing device shall be not less than 1 inch nor more than 1-1/2 inches. After firing the blasting cap the firing device shall be examined to determine conformance to the requirement of 3.5.5.2.

4.4.3.1.2 Failure during periodic tests. - In the event of failure of any firing device during periodic testing, inspection and acceptance of subsequent firing devices shall be suspended until the reasons for failure and consequent rejection have been determined and the cause of failure corrected by the contractor to the satisfaction of the procuring activity. The contractor shall furnish the procuring activity with a written statement properly endorsed by the Government inspector indicating the reasons for failure and the corrective action taken.



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4.4.3.2 Lot acceptance tests. -

4.4.3.2.1 Firing pin. -

4.4.3.2.1.1 Firing pin release. - The firing pin shall be tested for the force required to release it from its installed position in the firing device to determine conformance to 3.5.1.1. The safety pin and diaphragm shall be removed from the firing device and a gradually increasing force applied on the base of the firing pin. Suitable means shall be provided for measuring the applied force. The applied force shall be observed and noted at time of release of the firing pin.

4.4.3.2.1.2 Firing pin retention. - The firing pin shall be tested for retention in its installed position in the firing device to determine conformance to 3.5.1.2. The safety pin and the dummy base, if installed, shall be removed from the firing device. The firing device with the open end of the barrel in a downward position shall be shaken and jarred several times by hand and observation made to determine whether or not the firing pin has been thrown out of the firing device or become dislodged from its installed position.

4.4.3.2.2 Safety pin. - The safety pin shall be withdrawn from the firing device by means of a suitable load measuring device and observation made of the force required to withdraw the pin to determine conformance to 3.5.2.

4.4.3.2.3 Diaphragm. - The diaphragm shall be tested to determine conformance to 3.5.3. A suitable load measuring device shall be placed on the center of the "as installed" diaphragm and the load increased until the diaphragm snaps. The load shall be noted at the time the diaphragm snaps. The load measuring device shall then be placed on the center of the snapped diaphragm and the load increased until the diaphragm snaps back to its "as installed" height. The load shall be noted at the time the diaphragm snaps. The height of the diaphragm shall be measured before and after snapping. Test sample diaphragms, due to having been snapped, shall be scrapped as indicated on the applicable drawing and replaced in the lot from which they were taken by new and unsnapped diaphragms.

4.4.3.2.4 Leakage. - The firing device shall be tested for leakage to determine conformance to 3.5.4. An air line shall be connected to the base of the firing device by means of a suitable special fitting so as to apply internal air pressure within the firing device. The firing device shall be immersed in water or alcohol and subjected to an internal air pressure of not less than 5 pounds per square inch (psi) for a period of not less than 15 seconds and observation made to determine whether or not the firing device leaks air. The water or alcohol



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shall be clear and shall be motionless during the test. Water or alcohol shall not be permitted to enter the firing device and after completion of the test the firing device shall be thoroughly dry before removing the special fitting and installing the dummy base.

4.5 Resubmission. - Firing devices that fail to pass any one of the lot acceptance tests herein shall be considered defective. Rejected firing device lots may be resubmitted for retests in accordance with the procedures of Publication OP 400.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. -

5.1.1.1 Cleaning, drying, and preservation. - Not applicable.

5.1.1.2 Unit packaging. - Firing devices shall be packaged in accordance with Drawing LD 414462.

5.1.2 Level B and level C. - Not applicable.

5.2 Packing. -

5.2.1 Level A. - Firing devices packaged as specified in 5.1.1.2 shall be packed for shipment in accordance with Drawing LD 416389.

5.2.2 Level B and level C. - Not applicable.

5.3 Marking. -

5.3.1 For production firing devices. - In addition to any special marking required by the contract or order (see 6.2), inner containers and shipping containers shall be marked in accordance with Drawings LD 414462, LD 416389 and Standard MIL-STD-129 as applicable.

5.3.2 For preproduction firing devices. - Inner containers and shipping containers shall be marked as specified in 5.3.1. In addition, shipping containers shall be marked with the following information:

Title, number, and date of this specification
Mark and Mod and LD number
Sample for preproduction inspection
Name of manufacturer
Manufacturer's plant address
Reference (contract number)



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5.3.3 For firing devices for periodic testing. - Inner containers and shipping containers shall be marked as specified in 5.3.2 except that shipping containers shall be marked "Sample for periodic tests" instead of "Sample for preproduction inspection." In addition, shipping containers shall be marked with the identification of the lot or quantity from which the samples were taken.

6. NOTES

6.1 Intended use. - The Demolition Firing Device Mark 6 Mod 0 is intended for sympathetic detonation of explosive charges either in air or under water. When used under water the firing device is intended to be used at depths not exceeding 12 feet.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether preproduction samples and inspection are required (see 3.2 and 4.3.1.1).
(c) Testing activity for preproduction samples or samples for periodic tests if the activity is not designated by the Bureau of Ordnance (see 4.3.1 or 4.3.2).
(d) Frequency of sampling for periodic tests (see 4.3.2).
(e) Sampling for lot acceptance inspection if different from 4.3.3.
(f) Special marking when required (see 5.3.1).

6.3 General safety precautions. - The loading, assembly, and handling of the item covered by this specification, and the subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification will not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are certain minimum provisions which a contractor or manufacturer (which explosively loads the item covered) should observe in order to fulfill his responsibility for safety. At Bureau of Ordnance, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail technical requirements of the specification.

6.3.1 All loading operations should be conducted in a neat and orderly manner.



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6.3.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.3.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning, should insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.3.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.3.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with Intraplant distances when outside of loading rooms. For Navy managed explosives loading plants, the provisions of the Armed Services Explosives Safety Board covering quantity-distance relations for explosives will apply.

6.3.6 Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the precautionary measures in the following paragraphs should be observed.

6.3.6.1 Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

6.3.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operations.

6.3.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.3.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and,



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where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, low-energy type electric primers, detonators, and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high-frequency heat apparatus, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electrostatic effects but without inducing excessive moisture absorption by any of the components of the items being loaded.

6.3.6.5 Protect all explosive operations from effects of electrical current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed for example, from a defective electric soldering iron through a grounded contact. All electric type primers, detonators, or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting slips, or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.

6.3.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.



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6.3.6.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling, and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.3.6.8 Enforce good housekeeping and maintain effective policing, inspection, and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



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